

Sales Analysis & Performance Tracking System

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Flask	SQLAlchemy	Bootstrap 5 RTL	Chart.js	Arabic + English	Cloud Ready
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9	2,436	100	13
Employees ██████████	Sale Records ██████████ ██████████████	Pending Invoices ██████████ ██████████	Features █████ ██████████

Sahs Pro is a complete bilingual (Arabic/English) web application for sales analysis and performance tracking, designed specifically for Magrabi KSA retail branches. It processes XML-format Excel reports from SAP, tracks individual employee performance against monthly targets, and provides an intelligent assistant for instant insights.

Technical Architecture

Layer	Technology	Purpose
Backend	Python 3.11 + Flask	REST API, blueprints, session management
Database	SQLAlchemy + SQLite / PostgreSQL	ORM with migration support
Frontend	Bootstrap 5 RTL + Chart.js 4	Responsive, bilingual UI
i18n	Flask-Babel	Arabic (RTL) & English translations
Excel	XML parser (lxml/ET)	Handles SAP XML-format .xls files
Email	Flask-Mail (SMTP)	Send reports to recipients
Deploy	Gunicorn + Docker + Procfile	Cloud-ready (Heroku, Railway, Render)

Core Features | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

1

Excel Upload & Processing | 🔒🔒🔒🔒🔒🔒🔒🔒 Excel

Supports 3 SAP XML report types: Sales Productivity, Sales Detail, Pending Invoices. Auto-detects file type, parses sparse XML columns, deduplicates on import.

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Employee Management | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Add, edit, lock employees. Alias system prevents duplicate names from SAP. SAP ID linking for precise identification.

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Live Dashboard | 🔒🔒🔒🔒🔒🔒🔒🔒

Real-time KPIs: monthly sales, achievement %, top performer. Per-employee progress bars with motivational banners (green = ahead, yellow = behind).

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Monthly Targets | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Set company total target and distribute by % per employee. System calculates daily target = (target - achieved) / remaining working days.

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Actual vs Target Report | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Color-coded achievement table: green $\geq 100\%$, yellow $\geq 80\%$, red $< 80\%$. Variance column shows SAR difference from target.

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Weekly Comparison | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Current week vs previous week with trend arrows ($\uparrow\downarrow$). Automatic motivational messages based on performance change.

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Pending Invoices | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Track open invoices per employee. Add notes with reason codes (payment delay, dispute, missing docs, returned goods).

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Sales Advice Engine | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Rule-based alerts: employees under 80% for 2+ months, declining product categories, top performer highlighting.

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Sahs Pro Assistant | 🔒🔒🔒🔒🔒🔒🔒🔒

Chat widget answers natural language queries: "Who is the top seller?", "What is my remaining target today?", "Show pending invoices for employee X."

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Shift Schedule | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Monthly calendar view per employee. Morning/Evening/Day Off/Sick/Annual shifts with color coding. Calculates working days for target distribution.

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Export & Email | 🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒🔒

Export any report to Excel (openpyxl). Print-ready CSS. Email reports as attachments via SMTP (Gmail, Outlook, etc).

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Bilingual RTL/LTR | 🔒🔒🔒🔒🔒🔒🔒🔒

Full Arabic RTL and English LTR switching from navbar. Bootstrap 5 RTL CSS, Cairo font for Arabic, Inter for English.

Database Schema | [?] [?] [?] [?] [?] [?] [?] [?] [?] [?] [?]

Table	Description
employees	Employees master list with SAP ID and alias support
employee_aliases	Alternative name mappings to prevent duplicates
sale_records	2,436 detailed invoice lines per employee per date
sales_productivity	Aggregated qty/amount per category per period
sales_targets	Monthly targets per employee (amount + percentage)
company_targets	Company-level total monthly targets
pending_invoices	100 open invoices with employee links
invoice_notes	Employee comments on pending invoices
schedules	Daily shift assignments per employee
upload_batches	Audit log of all file imports
app_settings	Configurable system settings
saved_filters	User-saved custom filter combinations

API Endpoints

Method	Endpoint	Description
GET	/api/dashboard/kpis	Live KPI metrics (employees, sales, achievement)
GET	/api/employee/<id>/yearly-sales	Monthly totals for Chart.js line chart
GET	/api/employee/<id>/category-breakdown	Product category pie chart data
GET	/api/targets/chart	Target vs achieved per employee (bar chart)
POST	/uploads/upload	Upload and import Excel file (multipart)
POST	/assistant/chat	Ask Sahs Pro a question (JSON)
POST	/targets/set	Save monthly targets with percentages
GET	/reports/export/<type>	Download report as Excel file

Deployment Guide | [?????](#) [??????](#)

Local Development

```
$ git clone https://github.com/s7saiaagent-glitch/vibemap-server
$ git checkout claude/sales-analysis-bilingual-app-bfhiY
$ pip install -r requirements.txt
$ cp .env.example .env # edit your settings
$ python wsgi.py # open http://localhost:5000
```

Docker

```
$ docker build -t sahs-pro .
$ docker run -p 5000:5000 --env-file .env sahs-pro
```

Heroku / Railway / Render

```
$ Push code to GitHub
$ Connect repo to Heroku / Railway / Render
$ Add environment variables from .env.example
$ Deploy — the Procfile handles everything automatically
```

Environment Variables

Variable	Default	Description
SECRET_KEY	(required)	Flask secret key for sessions
DATABASE_URL	sqlite:///instance/sales.db	Database connection string
MAIL_SERVER	smtp.gmail.com	SMTP server for email reports
MAIL_PORT	587	SMTP port
MAIL_USERNAME		Email account username
MAIL_PASSWORD		Email app password
OPENAI_API_KEY	(optional)	For AI-enhanced assistant responses
WORKING_DAYS_PER_MONTH	26	Default working days for daily target calc